Linear Model – Performance and Error Analysis

1. Analysis objective

The model configuration, parameter search and final settings have already been documented. This analysis therefore focuses on understanding the behaviour of the final Linear Model rather than repeating its implementation.

The objective is to analyse how prediction accuracy changes across forecast horizons and circuits, identify where the largest errors occur, and highlight the main strengths and weaknesses that should later be considered during the final model comparison.

2. Validation and tuning stability

The parameter search produced a consistent result across all forecasting horizons. The same final configuration was selected for +1 h, +6 h and day-ahead forecasting:

- Pooled design without circuit identity
- LAD loss
- Alpha = 0.001

The validation scaled MAE across all 16 circuits was:

- +1 h: 0.340
- +6 h: 0.392
- Day-ahead: 0.395

This consistency is important because it shows that the selected configuration remains appropriate when the forecasting horizon changes.

The tuning results also show that increasing LAD regularisation beyond the selected value reduces performance. For example, at day-ahead, the validation score increases from 0.395 with alpha = 0.001 to 0.407 with alpha = 0.01 and 0.525 with alpha = 0.1.

The selected configuration therefore appears stable within the tested parameter grid rather than being an arbitrary choice.

3. Performance across forecasting horizons

The test results show a clear deterioration when moving from short-term forecasting to longer horizons.

Overall scaled MAE across the 16 circuits:

- +1 h: 0.613
- +6 h: 0.831

- Day-ahead: 0.838

Reported mean MAE across circuits:

- +1 h: 24.8 Wh
- +6 h: 38.5 Wh
- Day-ahead: 39.0 Wh

The main decrease in accuracy therefore occurs between +1 h and +6 h. The difference between +6 h and day-ahead is much smaller.

This suggests that the Linear Model benefits significantly from recent consumption information for very short-term forecasting. When the prediction moves further into the future, the immediate past becomes less informative and the model loses accuracy.

However, the limited difference between +6 h and day-ahead suggests that the model does not continue to deteriorate strongly once it reaches the longer forecasting range.

4. Performance by circuit

The overall results hide important differences between individual circuits.

At +1 h, several circuits are predicted with relatively low errors:

- NE|Plugs & Sockets: MAE = 4.58 Wh, scaled MAE = 0.024
- NW|RACK: MAE = 4.30 Wh, scaled MAE = 0.023
- NW|HVAC: MAE = 16.68 Wh, scaled MAE = 0.134

Other circuits generate substantially larger errors:

- NW|Lighting: MAE = 30.86 Wh, scaled MAE = 0.657
- NW|Plugs & Sockets: MAE = 175.60 Wh, scaled MAE = 0.312
- NW|Hand Dryer: MAE = 0.55 Wh, but scaled MAE = 0.624

This demonstrates why MAE and scaled MAE must be considered together.

For example, NW|Hand Dryer has a very small absolute error because the circuit itself consumes very little energy. Relative to its normal consumption level, however, the error remains significant.

Conversely, NW|Plugs & Sockets has a much higher MAE in Wh because it is a much larger consumer. Its relative error is less extreme once consumption scale is taken into account.

The model should therefore not be judged only through a global MAE. Forecasting performance is strongly dependent on the behaviour and consumption level of each circuit.

5. Effect of forecast horizon by circuit

The forecast horizon does not affect every circuit equally.

Some circuits remain relatively stable across all three horizons.

For NE|Plugs & Sockets:

- +1 h: 4.58 Wh
- +6 h: 3.62 Wh
- Day-ahead: 3.84 Wh

For NW|RACK:

- +1 h: 4.30 Wh
- +6 h: 3.20 Wh
- Day-ahead: 3.33 Wh

These circuits remain predictable even when the forecasting horizon increases.

Other circuits show a much stronger deterioration.

For NW|Lighting:

- +1 h: 30.86 Wh
- +6 h: 50.80 Wh
- Day-ahead: 50.51 Wh

For NW|Plugs & Sockets:

- +1 h: 175.60 Wh
- +6 h: 255.63 Wh
- Day-ahead: 257.04 Wh

This shows that forecasting difficulty depends on both the forecast horizon and the consumption behaviour of the circuit.

The Linear Model appears capable of maintaining relatively stable forecasts for regular circuits, while more variable loads become significantly harder to predict over longer horizons.

6. Large errors and abnormal operating behaviour

One of the clearest examples of the Linear Model's behaviour during an unusual event occurs for NE|HVAC on 3 October.

Before the event, actual consumption and predictions remain close to only a few Wh. Consumption then suddenly reaches 690 Wh while the model predicts only 9.83 Wh.

One hour later, actual consumption remains at 686 Wh and the model has reacted, predicting approximately 594.87 Wh.

The following prediction remains high at approximately 590.92 Wh against an actual value of 266 Wh. The model then predicts approximately 226.39 Wh after actual consumption has already returned to only 3 Wh.

This sequence highlights three characteristics of the model:

1. The Linear Model struggles to anticipate a sudden operating change that is not represented in recent historical behaviour.
2. Once the change enters the historical input variables, the model reacts strongly.
3. The influence of the unusual event can remain in subsequent predictions after actual consumption has already returned to normal.

This behaviour is consistent with the model's strong reliance on historical variables such as recent consumption, 24-hour statistics, 168-hour statistics and lagged energy values.

The important limitation is therefore not permanently poor accuracy, but sensitivity to sudden changes in operating conditions.

7. MAE and RMSE: sensitivity to extreme errors

The difference between MAE and RMSE provides additional information about the model's error distribution.

For NE|HVAC at +1 h:

- MAE = 8.77 Wh
- RMSE = 61.08 Wh

For NW|Emerg. Lighting:

- MAE = 24.69 Wh
- RMSE = 79.51 Wh

The large difference between MAE and RMSE indicates that some individual predictions produce very large errors.

In contrast, NE|Plugs & Sockets has:

- MAE = 4.58 Wh
- RMSE = 6.10 Wh

and NW|RACK has:

- MAE = 4.30 Wh
- RMSE = 5.74 Wh

For these circuits, the difference is much smaller, indicating a more regular error pattern without the same influence from isolated extreme deviations.

This distinction is important for the final evaluation:

- MAE represents the typical size of the forecasting error.
- RMSE helps reveal whether the model occasionally produces very large errors.

The Linear Model therefore appears reliable on some regular circuits but sensitive to extreme events on specific electrical loads.

8. Behaviour on circuits outside the training set

The Linear Model was trained on the eight eligible NE/NW circuits but produces predictions for all 16 circuits.

The selected design does not include circuit identity, meaning predictions are mainly driven by the consumption history and temporal characteristics of each circuit rather than a learned circuit-specific label.

The model can therefore produce forecasts for SE/SW circuits that were not included during model fitting.

However, these results must be interpreted carefully because the available test coverage is substantially lower for several SE/SW circuits.

Examples at +1 h include:

- SE|HVAC: 38.1% coverage
- SW|Emerg. Lighting: 34.5%
- SW|Hand Dryer: 46.4%
- SW|Plugs & Sockets: 56.5%

The results demonstrate that the model is technically capable of transferring its learnt relationships to previously unseen circuits. However, incomplete target coverage means that there is less evidence available to assess the reliability of these predictions.

For this reason, performance on low-coverage SE/SW circuits should not receive the same interpretative weight as results obtained on the fully covered NE/NW circuits.

9. Hour-of-day analysis

An important additional dimension is whether the model performs differently depending on the hour of the day.

The prediction results contain the actual consumption, predicted consumption and local Madrid target hour. This allows the absolute forecasting error to be calculated as:

Absolute Error = |Actual Wh - Predicted Wh|

The errors can then be grouped by local hour:

- 00:00
- 01:00
- 02:00
- …
- 23:00

and analysed separately for +1 h, +6 h and day-ahead forecasts.

The individual records already demonstrate that specific operating events can create very large errors during particular hours. However, the available analysis does not yet provide sufficient evidence to conclude that a specific hour is systematically the weakest period across all circuits.

A complete hour-of-day analysis should therefore distinguish between:

- recurring errors associated with particular operating hours;
- circuit-specific patterns;
- isolated abnormal events.

This will make it possible to determine whether the Linear Model has a systematic time-of-day weakness or whether the largest hourly errors are mainly caused by unusual operating events.

10. Model-specific limitations

The main limitations identified specifically for the Linear Model are:

Sensitivity to abrupt operating changes

The NE|HVAC event shows that the model can initially miss a sudden consumption change, react strongly once the change appears in its historical inputs, and continue to reflect part of that change after actual consumption has returned to normal.

Strong performance heterogeneity between circuits

The forecasting quality varies considerably depending on the electrical circuit. Global metrics therefore hide important differences in model behaviour.

Uneven evaluation coverage for unseen circuits

Several SE/SW circuits contain significantly fewer valid evaluation hours. Predictions can be generated for these circuits, but the available evidence is insufficient to assess them with the same confidence as fully covered circuits.

Dependence on historical consumption patterns

The model relies heavily on lagged consumption and historical statistics. This works well when consumption follows relatively stable patterns but becomes less reliable when operating behaviour changes suddenly.

No external contextual information

The current Linear Model does not use weather variables or the UPM academic calendar. Its predictions are mainly based on historical consumption and temporal features.

11. Key Results for Final Model Comparison

The following results summarise the main information that should later be used to compare the Linear Model with the other candidate models.

Final configuration

- Model type: LAD Linear Model
- Design: pooled
- Circuit identity: not used
- Alpha: 0.001
- Same configuration selected across all three horizons

Validation scaled MAE

- +1 h: 0.340
- +6 h: 0.392
- Day-ahead: 0.395

Test overall scaled MAE

- +1 h: 0.613
- +6 h: 0.831
- Day-ahead: 0.838

Mean test MAE across circuits

- +1 h: 24.8 Wh
- +6 h: 38.5 Wh
- Day-ahead: 39.0 Wh

Strongest horizon

+1 h.

The largest overall loss of accuracy occurs between +1 h and +6 h. Performance changes only slightly between +6 h and day-ahead.

Strong circuit behaviour

NE|Plugs & Sockets and NW|RACK remain accurately predicted across all three horizons, indicating that the model performs well on relatively stable consumption profiles.

Main difficult cases

More variable circuits such as NW|Lighting and NW|Plugs & Sockets show a stronger degradation at longer horizons.

Abnormal operating events can also create large individual errors, as demonstrated by NE|HVAC.

Extreme-error sensitivity

Large differences between MAE and RMSE for NE|HVAC and NW|Emerg. Lighting indicate sensitivity to occasional extreme prediction errors.

Generalisation

The model can produce forecasts for circuits outside the training set because the selected design does not depend on circuit identity. However, low SE/SW test coverage limits the strength of conclusions regarding this generalisation.

Contextual information

Weather and academic-calendar information are not used in the current version.

12. Conclusion

The Linear Model provides a stable forecasting approach, with the same LAD configuration (alpha = 0.001, pooled design without circuit identity) selected across +1 h, +6 h and day-ahead forecasting.

Its clearest strength is short-term forecasting. The overall test scaled MAE increases from 0.613 at +1 h to 0.831 at +6 h and 0.838 day-ahead, indicating that most of the loss of accuracy occurs when moving beyond the immediate short-term horizon.

Performance is highly dependent on the circuit. Stable consumption patterns such as NE|Plugs & Sockets and NW|RACK are predicted consistently across horizons, while more variable circuits become substantially more difficult at longer horizons.

The main weakness identified is sensitivity to abrupt operating changes. The NE|HVAC event demonstrates that the model can initially fail to anticipate a sudden change, adapt after observing it, and then temporarily preserve its influence after the actual consumption has returned to normal. The large differences between MAE and RMSE on several circuits confirm that occasional extreme errors remain an important weakness.

For the final model comparison, the Linear Model can therefore be summarised as follows:

Strengths: stable parameter selection, strong +1 h performance, good accuracy on regular consumption profiles, ability to generate predictions for unseen circuits, and relatively consistent behaviour for several stable electrical loads.

Weaknesses: lower accuracy at longer horizons, large differences in performance between circuits, sensitivity to abrupt consumption changes and occasional high-error events, and limited evidence on several SE/SW circuits because of incomplete coverage.

Overall assessment: the Linear Model is particularly suited to short-term forecasting of relatively stable energy-consumption patterns. Its relevance for longer horizons and highly variable circuits is weaker, mainly because a linear relationship based on historical behaviour cannot always adapt rapidly to sudden changes in building operation.

This summary provides the main elements to carry forward into the final comparison: MAE/RMSE performance, performance by horizon, circuit-level consistency, response to abnormal events, and robustness of the forecasts.